Taaza News → Excel Sync

Project details, architecture and run report

Project folder: `wordtoexcel\word_to_excel_sync` · Report date: 13 October 2026 · Latest commit: `bbfd55c` "Add Windows 7 SP1 64-bit build"

157 / 157
unit tests passed (3.1 s)

4
news feeds synced to 4 Excel workbooks

~4 s
from saving a .txt file to the workbook updating

Win 7 – 11
separate builds for Windows 7 SP1 and 10/11

1. What the project is

Taaza News Excel Sync is a Windows desktop app for a Hindi news channel's playout workflow. Editors write news lines in Windows Notepad `.txt` files. The playout (on-air graphics) software reads from Excel workbooks. This app connects the two: it watches four specific `.txt` files and keeps four Excel workbooks as **live mirrors** of them. Each workbook uses the exact sheet names, columns, fonts and formats that the playout software expects.

It runs 24/7 in the background. It starts monitoring as soon as it opens, picks up changes made while it was closed, strips `<BITMAP>…</BITMAP>` image placeholders, and preserves Hindi text exactly, with no translation and no rewording.

Notepad .txt row_01 / 04 / 09 / 10 → **Watcher** watchdog, debounced 1.5 s → **TXT reader** encoding fallback, BITMAP removal → **SQLite cache** sync.db, hash check → **Excel manager** atomic rebuild of workbook → **Playout software** reads .xlsx

2. The four feeds

Feed	Source file	Destination workbook	Sheet(s)	Columns
Top News	`row_01.txt`	Top News LT.xlsx	Top News LT	Status, News
Upper Band	`row_10.txt`	Upper Band.xlsx	Breaking News FF	Visibility, News
Breaking News	`row_04.txt`	Breaking LT.xlsx	Breaking LT News + Breaking LT Category	Visibility, Category, News
Full Screen Breaking	`row_09.txt`	Breaking FF + Top.xlsx	Breaking News FF	Visibility, News

Every other file in the TXT folder (for example `row_02.txt`) and every other workbook in the Excel folder is ignored and never touched. Source and destination file names can be changed in the GUI under **FEED MAPPING**, which is password-protected. Sheet names and column layouts are fixed in code because the playout software matches on them.

Defaults filled in automatically

Field	Default	Rule
Visibility / Status	1 (visible)	Every row gets the value; 0 would mean hidden.
Category (Breaking LT)	08:00:00 PM	First 2 news lines.
Category (Breaking LT)	09:00:00 PM	Line 3 onward, including lines added later. The 8 PM block never shifts.

3. Run report

3.1 Automated tests

The suite was run with Python 3.14.7 and `python -m unittest discover -s tests`. Result: **157 tests, all passed, in 3.1 s**. The console printed some "Excel locked" and "config could not be read" warnings. These come from tests that deliberately exercise error handling, not from failures.

Test file	Covers
`test_txt_reader.py`	BITMAP removal, encoding fallback (UTF-8/BOM, UTF-16, cp1252, latin-1), line extraction
`test_excel.py`	Exact workbook layouts, fonts, widths, time formats, locked-file handling
`test_sync.py`	Create / modify / delete / recreate per feed; strict file filtering
`test_feeds.py, test_config.py`	Feed definitions, category-time parsing, config load/save (including BOM)
`test_database.py, test_hindi_detector.py`	SQLite cache, Devanagari helpers

3.2 Live run of the application

The app's own `config.json` points at the live playout folder `Desktop\Excel taza tv`. So that the run would not overwrite on-air workbooks, the GUI was launched against a **sandbox copy** of `demo_data` (eleven files, `row_01` – `row_11.txt`) with a separate scratch Excel folder. Results:

- **Startup:** the initial scan found all four feeds and wrote four workbooks. Status was RUNNING with 4 feeds synced, 21 news items and no Excel writes pending.
- **Filtering:** the seven non-feed files (`row_02, 03, 05–08, 11`) were ignored.
- **BITMAP removal:** every `<BITMAP>` tag was stripped, and lines that held only a tag were dropped.
- **Live watch:** a new Hindi line was appended to `row_04.txt` while the app ran. `Breaking LT.xlsx` was rebuilt about 4 seconds later with 5 news rows, and the new line correctly got 09:00:00 PM.

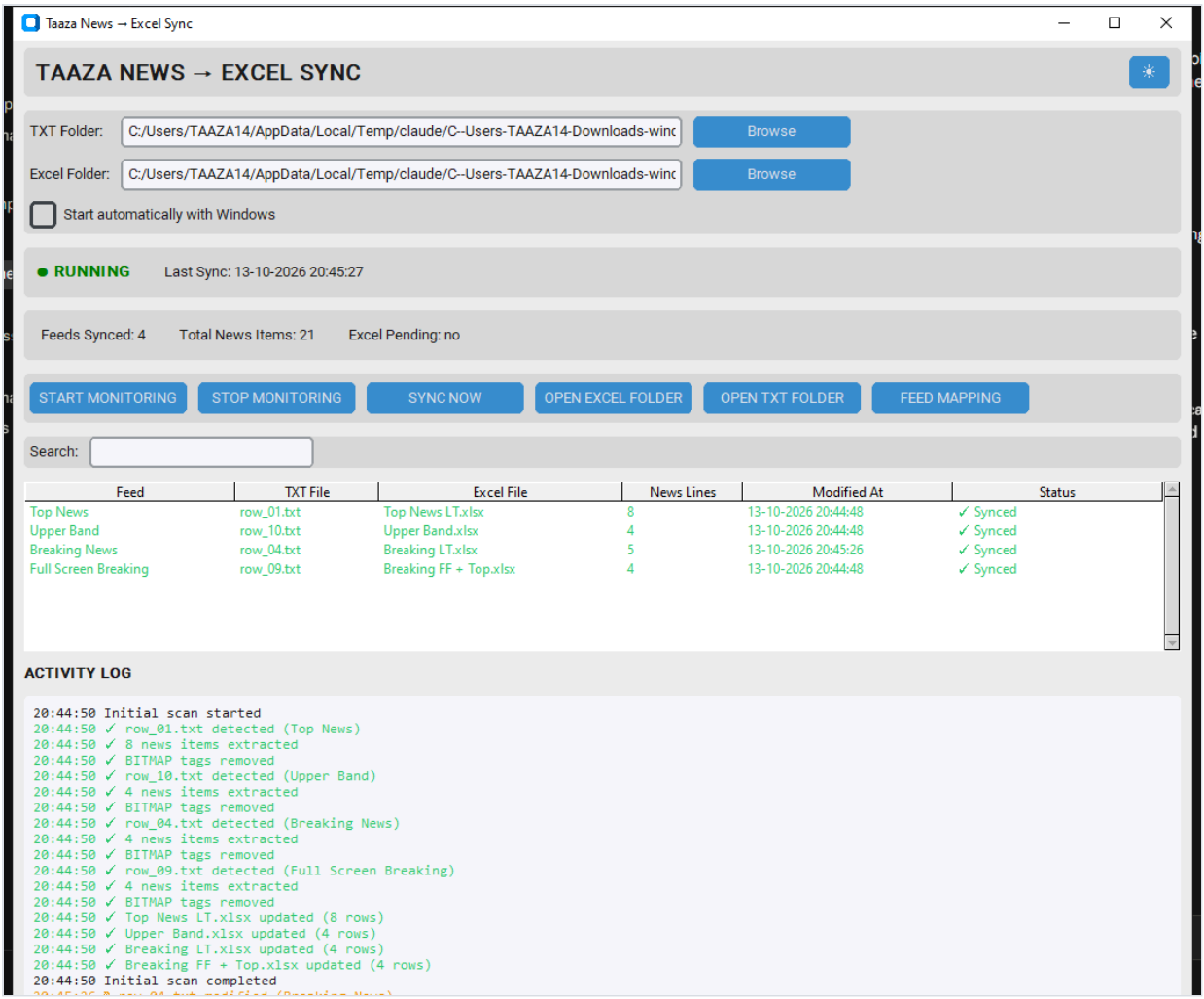


Figure 1: The running application (sandbox data) after the initial scan and one live edit to row_04.txt.

3.3 Generated Excel output (from the live run)

Top News LT.xlsx → sheet *Top News LT*

Status	News
1	राष्ट्रीय मंदिर के पास इलाके में भीषण आग
1	A major fire occurred near the temple.
1	दर्शनार्थियों की भीड़ में अचानक मौत
1	The crowd suddenly faced a tragedy.
1	7 श्रद्धालुओं की मौत, 12 से अधिक लोग घायल
1	Seven people died and many were injured.
1	भारत ने India के साथ नई बातचीत शुरू की
1	India started a new discussion.

Source line राष्ट्रीय मंदिर के पास इलाके में भीषण आग `<BITMAP>:\newsair\logo.psd</BITMAP>` became the first row above, with the BITMAP tag removed. The file's last line held only a BITMAP tag, so it was dropped.

Breaking LT.xlsx → sheet *Breaking LT News* (after the live edit)

Visibility	Category	News
1	08:00:00 PM	अभी-अभी: मुख्यमंत्री ने बड़ी घोषणा की
1	08:00:00 PM	Just in: the Chief Minister made a major announcement.
1	09:00:00 PM	देशभर में सुरक्षा व्यवस्था कड़ी की गई
1	09:00:00 PM	Security has been tightened nationwide.
1	09:00:00 PM	लाइव टेस्ट: नई ब्रेकिंग खबर जोड़ी गई *(added live)*

Breaking LT.xlsx → sheet *Breaking LT Category*: Visibility 1 / 08:00:00 PM and Visibility 1 / 09:00:00 PM. This sheet lists each distinct time once, taken from the data.

Upper Band.xlsx and **Breaking FF + Top.xlsx** → sheet *Breaking News FF*

Workbook	Visibility	News
Upper Band	1	अपर बैंड: शेयर बाजार में उतार-चढ़ाव जारी
Upper Band	1	सोने-चांदी के दाम में मामूली बदलाव
Breaking FF + Top	1	पूरी स्क्रीन ब्रेकिंग: बड़ा फैसला सुनाया गया
Breaking FF + Top	1	सभी चैनलों पर लाइव प्रसारण जारी

The English lines in each of these two files are omitted here for brevity; the workbooks contain them.

Excel formatting per layout

Layout	Header	Body font	Column widths	Extras
Top News LT	Bold, bright yellow, centred	Nirmala UI 11	A 13, B 100	Frozen at A2, AutoFilter
Breaking News FF	Yellow, not bold	Arial 10	B 68.71	No freeze, no filter
Breaking LT	Yellow	Arial 10	B 20.43, C 57	Category is a real Excel time, format `hh:mm:ss AM/PM`

4. Architecture

Module	Lines	Responsibility
`main.py`	76	Entry point: logging, single-instance guard, config location (next to `main.py`, or `%APPDATA%\TaazaNewsExcelSync` for the EXE)
`app/gui.py`	709	CustomTkinter UI: folders, status, buttons, feed list, activity log, Feed Mapping editor, day/night theme, 10 s background re-scan
`app/sync_engine.py`	295	Per-feed create/modify/delete orchestration, strict path filtering, pending-Excel retry
`app/excel_manager.py`	227	The three workbook layouts and safe atomic writes
`app/feeds.py`	200	Four feed definitions, formats, category-time parsing
`app/database.py`	160	SQLite cache of current source files, keyed by feed

`app/config.py`	152	Load/save `config.json` (BOM-safe); a load failure is reported loudly
`app/watcher.py`	130	watchdog observer, 1.5 s debounce, only the four file names
`app/txt_reader.py`	122	Encoding fallback chain, BITMAP removal, line extraction
`app/utils.py`	98	SHA-256 hashing, retries, file-size stability checks
`app/single_instance.py`	53	Named mutex: only one copy may run
`app/models.py`, `logger.py`, `scanner.py`, `errors.py`, `hindi_detector.py`	150	Data model, rotating log (5 × 2 MB), path resolution, shared errors, Devanagari helpers (unused by sync)

Change handling

Event	Behaviour
New file	Waits until the file size stabilises, cleans it, writes the feed's workbook
Modified file	Re-reads it, updates the DB record in place, rebuilds the workbook (no duplicate rows)
Deleted file	Removes the record; the workbook is rewritten with headers only. Other feeds are untouched
Rapid repeated events	Coalesced within 1.5 s, so one save is processed once
Workbook open in Excel	Marked "Excel Pending", retried about every 2 s until the file is closed
Missed events / app closed	Full reconciliation on startup, on SYNC NOW, on Browse / START MONITORING, and every 10 s in the background

Database (`sync.db`, table `feeds`)

`id, feed_key (UNIQUE), label, txt_file, full_path, content, file_hash (SHA-256), size, mtime, created_at, modified_at, last_synced_at`

The database is a cache, not a history. A row exists only while its source file exists. Unchanged files are skipped by checking size and mtime first, then the hash.

5. Running and building

Run from source

```
cd word_to_excel_sync
python -m pip install -r requirements.txt
python main.py
```

Dependencies: watchdog, openpyxl, customtkinter, plus pillow and pyinstaller for building only. Default folders: TXT `Documents\notepad`, Excel `Desktop\Excel taza tv`. Both can be changed with **Browse**.

Build outputs already present

Target	Build script	Output	Size
Windows 10/11	`build_exe.ps1`	`dist\TaazaNewsExcelSync.exe`	30.7 MB
Windows 10/11 installer	`installer.iss`	`installer_output\TXT_News_Excel_Sync_Setup.exe`	33.7 MB

Windows 7 SP1 64-bit	`build_win7.bat`	`dist-win7\TaazaNewsExcelSync.exe`	11.7 MB
Windows 7 installer	`installer_win7.iss`	`installer_output_win7\TaazaNewsExcelSync_Win7_Setup.exe`	37.6 MB
Portable	—	`TaazaNewsExcelSync_Portable.zip`	11.5 MB

The Windows 7 build is pinned to Python 3.8.10 and PyInstaller 5.13.2, with watchdog 4.0.2, openpyxl 3.1.5 and customtkinter 5.2.2. Python 3.8 is the last CPython release that supports Windows 7. The installer `python-3.8.10-amd64.exe` is kept in `wordtoexcel\win7_build`.

6. Findings and recommendations

Severity	Finding	Recommendation
Bug	In `app/gui.py` line 104, the window-close handler (`WM_DELETE_WINDOW` → `_on_close`) is registered inside `_warn_if_config_unreadable()`, after its early `return`. In normal use the handler is never registered, so closing the window skips stopping the watcher and closing the database.	Move the `self.protocol(...)` call into `__init__`.
Caution	The `config.json` in the project folder targets the live playout folder `C:\Users\TAAZA14\Desktop\Excel taza tv`. Running `python main.py` from source on this machine writes directly to the on-air workbooks (for example `Breaking LT.xlsx`).	For development or testing, point the folders at a test location, or keep a separate dev config.
Caution	The `venv-win7` folder is broken on this machine. It points at `F:\PyWin7Build\Python38\python.exe`, which does not exist.	Re-create it with the bundled Python 3.8.10 before running `build_win7.bat`.
Minor	The Feed Mapping password (default `2001`) is stored in plain text in `config.json`. It guards against accidental edits, not against determined users.	Change the default on production machines.
Info	`pytest` is not installed. The suite runs with the standard library `unittest`.	Optionally add pytest to a dev-requirements file.
Info	`hindi_detector.py` is unused by the sync path, as documented. System-tray support is intentionally absent.	None required.

7. Troubleshooting (quick reference)

Symptom	Cause / fix
A file never shows up	Only row_01, row_04, row_09 and row_10 are read. Check FEED MAPPING.
Workbook shows headers only	Its .txt file is missing or empty.
Workbook doesn't update	It is open in Excel. The app retries every ~2 s; close the workbook.
Wrong category time	Adjust Default Category or “First N news lines” in FEED MAPPING.
Second copy won't open	By design, only one instance may run.
Logs	`logs\app.log` (rotated, 5 × 2 MB)